

Run report

2026-09-10T132409Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	1,600,000
dim	12	dataset	jets
seed	2,113,982,479	data_seed	42
hidden_units	128	n_layers	3
n_epochs	100	n_disc_steps	5
lr_g	0.00003000	lr_d	0.0001000
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	0.0003705	best_epoch	38
compile_cache_warm	True		
variables	m, M, w, tau21, zg, sdm, q, f_ch, lha, ang2, ptd, n_ch		
mmd_sigmas_detector	$3.107 \times (1/2, 1/\sqrt{2}, 1, \sqrt{2}, 2)$		
mmd_sigmas_particle	$3.027 \times (1/2, 1/\sqrt{2}, 1, \sqrt{2}, 2)$		

Timing

Phase	Seconds	Share (%)	Detail
train	58.07	79.6	train
transfer	1.449		train
compile	6.796		train
epochs	48.06		train
select	1.500		train
particle_mmd	0.5562	0.763	train
save	0.07085	0.0971	train
data	3.008	4.12	cache hit – load
load	1.811	2.48	load
plots	9.413	12.9	load
evaluate	0.001440	0.00197	load
total	72.93	100	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Observable	Wasserstein					JS divergence					VLC divergence [$\times 10^3$]				
	Sim.	IBU	IBU impr. (%)	RAN	RAN impr. (%)	Sim.	IBU	IBU impr. (%)	RAN	RAN impr. (%)	Sim.	IBU	IBU impr. (%)	RAN	RAN impr. (%)
<i>Mass and hard scale (IRC-safe kinematics)</i>															
m [GeV]	0.2089	0.1388	33.6	0.01157	94.5	0.006762	0.003666	45.8	0.00009512	98.6	26.93	14.62	45.7	0.3599	98.7
$\ln \rho$	0.2279	0.02071	90.9	0.01322	94.2	0.008383	0.0004017	95.2	0.0001516	98.2	33.31	1.576	95.3	0.5864	98.2
<i>Continuous angularities (IRC-safe jet shapes)</i>															
$\lambda_{0.5}^1$	0.2626	0.01284	95.1	0.01161	95.6	0.009359	0.0004714	95.0	0.0001209	98.7	37.19	1.877	95.0	0.4777	98.7
w	0.2656	0.009978	96.2	0.01141	95.7	0.009139	0.0004527	95.0	0.0001261	98.6	36.32	1.800	95.0	0.4965	98.6
λ_2^1	0.2554	0.009215	96.4	0.0131	94.9	0.007776	0.0001728	97.8	0.0001294	98.3	30.90	0.6583	97.9	0.4798	98.4
<i>Splitting and 2-prong substructure</i>															
$z_g^\dagger$	0.06045	0.06045	0.0	0.01368	77.4	0.0006723	0.0006723	0.0	0.0001035	84.6	2.683	2.683	0.0	0.4140	84.6
$\tau_{21}^{(\beta=1)}$	0.1218	0.07010	42.4	0.01768	85.5	0.003372	0.001934	42.7	0.0001638	95.1	13.44	7.712	42.6	0.6504	95.2
<i>Hadronization, multiplicity and fragmentation (IRC-unsafe)</i>															
M	0.2242	0.1773	20.9	0.02168	90.3	0.009640	0.007633	20.8	0.0001830	98.1	38.25	30.31	20.8	0.7004	98.2
n_{ch}	0.2208	0.1723	22.0	0.02287	89.6	0.007512	0.006314	16.0	0.0001431	98.1	29.81	25.07	15.9	0.5459	98.2
f_{ch}	0.05286	0.04094	22.6	0.007417	86.0	0.001033	0.0007960	22.9	0.00009904	90.4	4.125	3.177	23.0	0.3961	90.4
p_T^D	0.2568	0.03080	88.0	0.01857	92.8	0.008223	0.001311	84.1	0.0001520	98.2	32.65	5.211	84.0	0.6035	98.2
q	0.07967	0.01751	78.0	0.007496	90.6	0.001576	0.0002512	84.1	0.00008472	94.6	6.261	0.9940	84.1	0.3292	94.7

[†] IBU's purity binning produced fewer than two bins for this observable, so IBU declined to unfold it and returned its input unchanged. The improvement shown for it is not a measurement.

Particle-level metrics

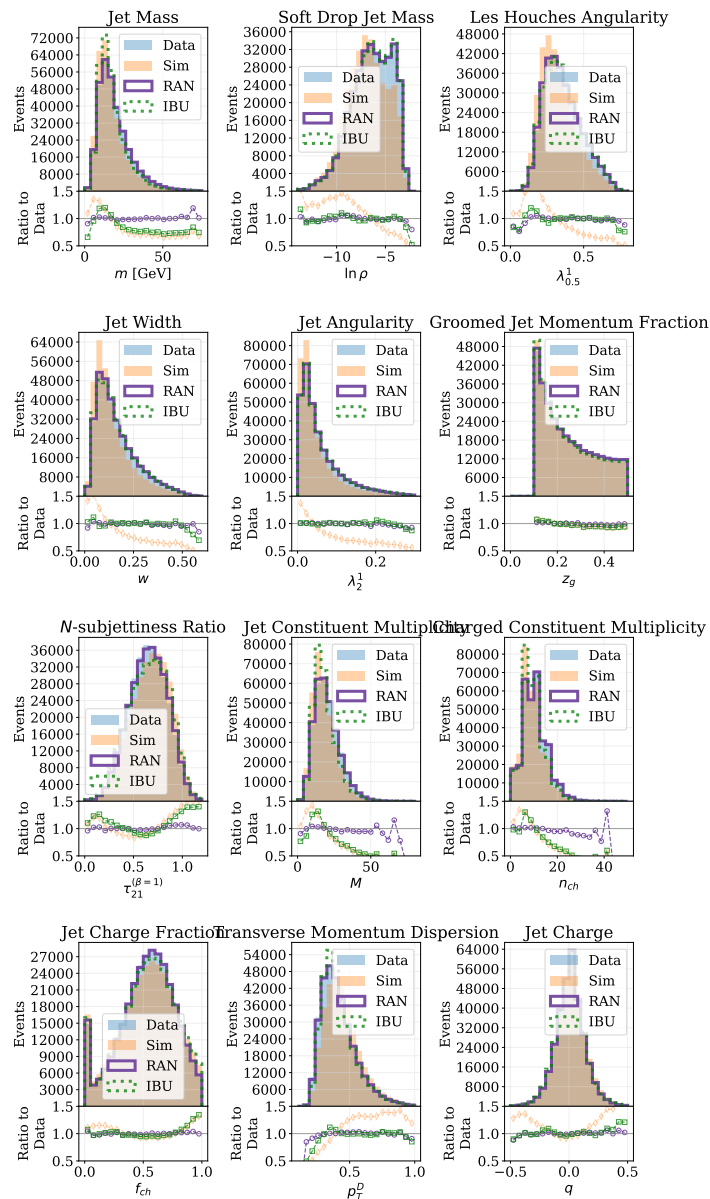
Distance to Truth. Improvements are relative to Generation.

Observable	Wasserstein					JS divergence					VLC divergence [$\times 10^3$]				
	Gen	IBU	IBU impr. (%)	RAN	RAN impr. (%)	Gen	IBU	IBU impr. (%)	RAN	RAN impr. (%)	Gen	IBU	IBU impr. (%)	RAN	RAN impr. (%)
<i>Mass and hard scale (IRC-safe kinematics)</i>															
m [GeV]	0.1117	0.09408	15.8	0.08980	19.6	0.001963	0.01186	−504.3	0.001424	27.5	7.809	45.93	−488.2	5.652	27.6
$\ln \rho$	0.2531	0.03257	87.1	0.01425	94.4	0.009878	0.001731	82.5	0.0001730	98.2	39.27	6.889	82.5	0.6779	98.3
<i>Continuous angularities (IRC-safe jet shapes)</i>															
$\lambda_{0.5}^1$	0.3030	0.01871	93.8	0.008400	97.2	0.01234	0.002009	83.7	0.0001850	98.5	48.98	8.014	83.6	0.7337	98.5
w	0.2841	0.01229	95.7	0.006554	97.7	0.01139	0.001465	87.1	0.0001999	98.2	45.23	5.837	87.1	0.7825	98.3
λ_2^1	0.2538	0.01113	95.6	0.006227	97.5	0.008824	0.0008335	90.6	0.0001384	98.4	35.08	3.315	90.5	0.5365	98.5
<i>Splitting and 2-prong substructure</i>															
$z_g^\dagger$	0.1134	0.1134	0.0	0.01512	86.7	0.002041	0.002041	0.0	0.0001317	93.6	8.086	8.086	0.0	0.5262	93.5
$\tau_{21}^{(\beta=1)}$	0.2592	0.1505	41.9	0.01042	96.0	0.01050	0.006577	37.4	0.0001377	98.7	41.75	26.17	37.3	0.5433	98.7
<i>Hadronization, multiplicity and fragmentation (IRC-unsafe)</i>															
M	0.2568	0.2048	20.2	0.02069	91.9	0.007842	0.01283	−63.7	0.0002267	97.1	31.10	49.85	−60.3	0.8651	97.2
n_{ch}	0.2346	0.1886	19.6	0.02269	90.3	0.006447	0.009773	−51.6	0.0001707	97.4	25.58	38.38	−50.1	0.6458	97.5
f_{ch}	0.04578	0.04533	1.0	0.009442	79.4	0.001025	0.0009435	7.9	0.0001305	87.3	4.095	3.766	8.0	0.5209	87.3
p_T^D	0.2363	0.05536	76.6	0.01685	92.9	0.008342	0.004029	51.7	0.0002556	96.9	33.05	16.05	51.4	1.011	96.9
q	0.07937	0.01815	77.1	0.007897	90.1	0.002027	0.0009958	50.9	0.0001020	95.0	7.982	3.921	50.9	0.3605	95.5

[†] IBU's purity binning produced fewer than two bins for this observable, so IBU declined to unfold it and returned its input unchanged. The improvement shown for it is not a measurement.

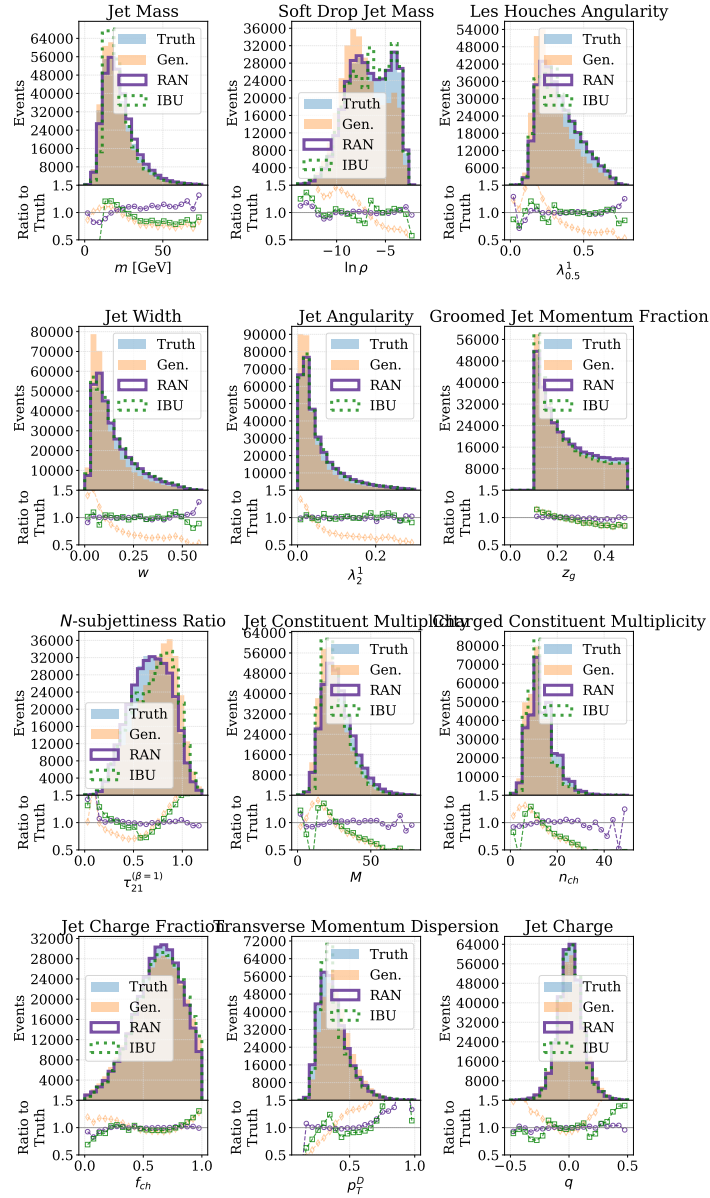
Detector-level distributions

Detector Level



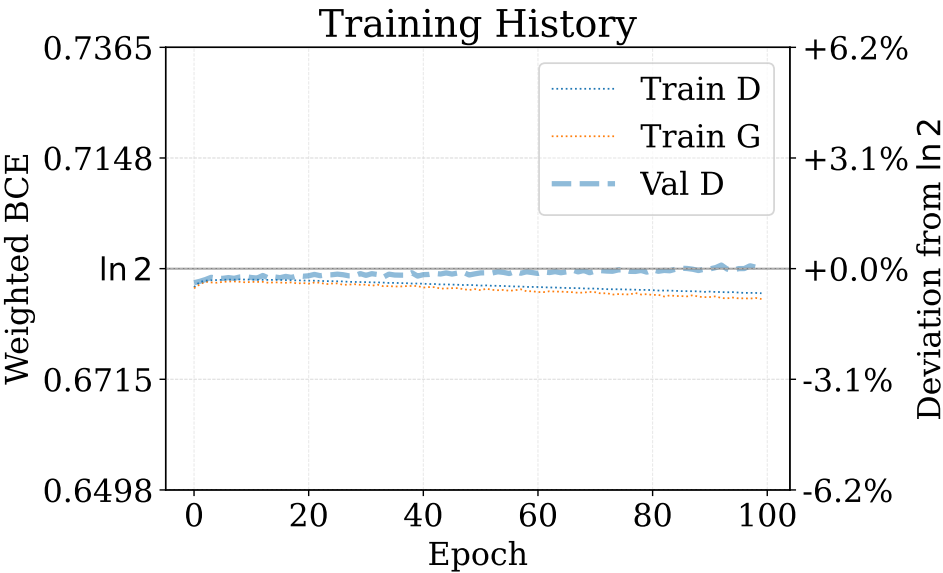
Particle-level distributions

Particle Level



Training diagnostics

Training losses



Selection criteria

